

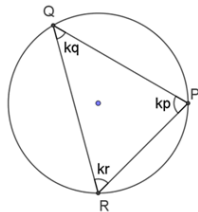
S1Q17

Question 17 – Stage 1

I roll three fair, six-sided dice with sides labelled 1 to 6 and the scores are p , q and r (which are not necessarily distinct).

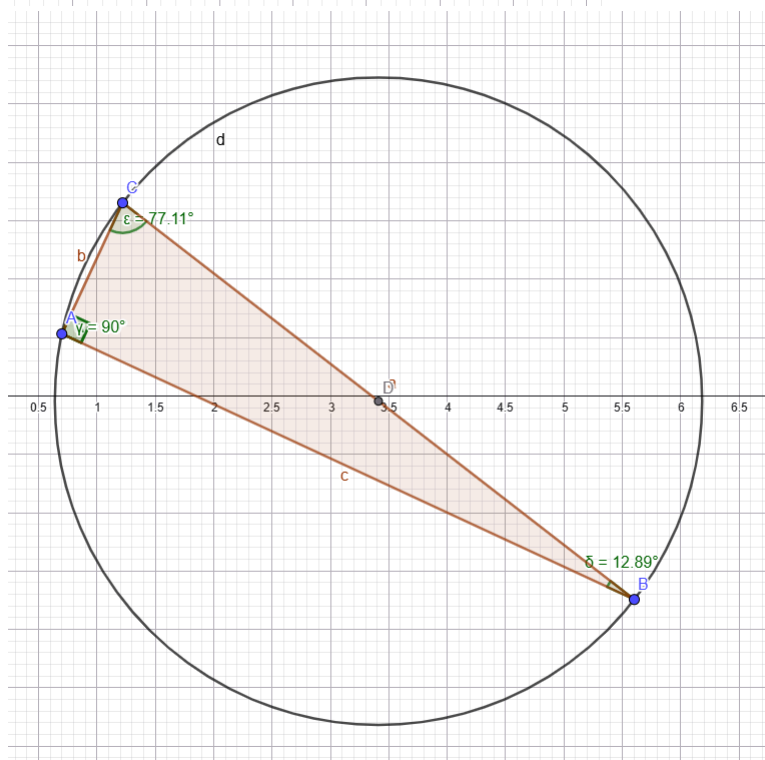
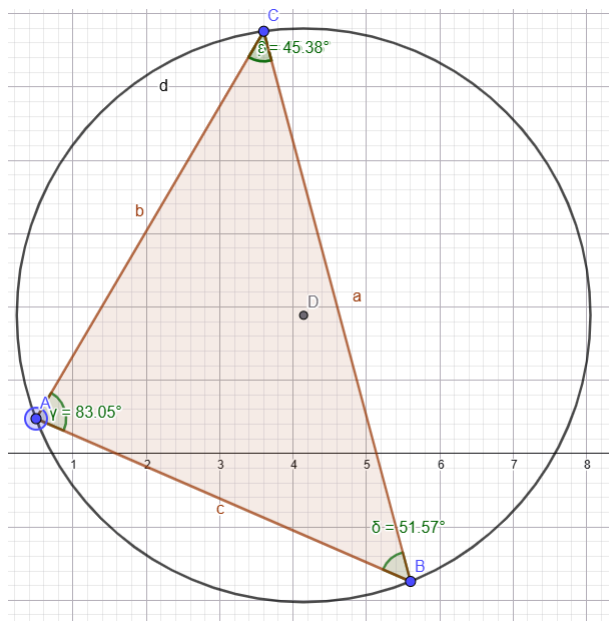
I construct a triangle PQR with vertices on a circle, and with interior angles in the ratio $p : q : r$.

What is the probability (to 3.d.p.) that the centre of the circle lies neither inside nor on an edge of the triangle?

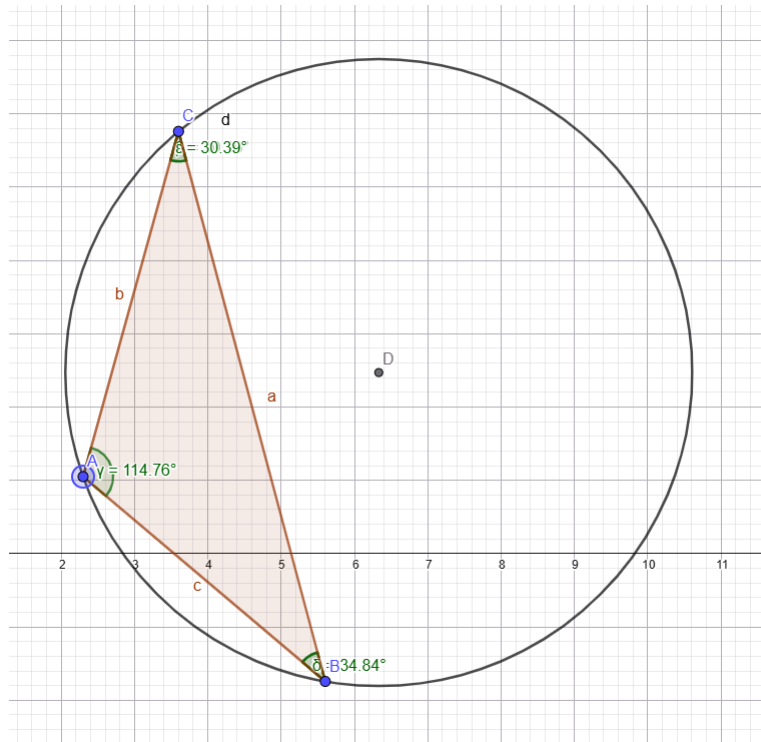


Working

When the sum of the two smaller angles is greater than or equal the largest angle, the centre is within the triangle or on its edge.



When the sum of the smaller two angle is less or equal to the largest angle, the centre is outside the triangle or on the edge.



This is because the angle at the centre of the circle between the two smaller angles will always be double the third angle. If the third angle is ≤ 90 the angle at the center will be ≤ 180 meaning the centre is inside the PQR or on its edge. If the third angle is > 90 the angle at the center will be > 180 meaning PQR will not contain the center of the circle.

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In [ ]: from itertools import product

countTotal = 0
countOutside = 0

for roll in product(range(1,7),repeat=3):
    countTotal += 1
    p,q,r = sorted(roll,reverse=True) # Ensure p is largest angle
    if p > q+r: # Since it is ratio, no values need to be converted into degrees
        countOutside+=1

probability = countOutside/countTotal
print(f"Probability: {round(probability,3)}")

Probability: 0.278
```